

TREADGUARD CV

Predictive Road Audits & Abrasion Analytics

Technical Manual & System Documentation

IIT Madras — Road Safety Hackathon 2026

Track: RoadWatch | Theme: Human-AI Collaboration

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Team Name: Impactious

Hackathon Listing: <https://unstop.com/hackathons/road-safety-hackathon-2026-iit-madras-1680515>

Trial Footage, G-Force CSVs & Sample Reports:

[\[Google Drive Folder \]](#)

Build 2026.05

About This Project

TreadGuard CV is a browser-based road intelligence platform built for the IIT Madras Road Safety Hackathon 2026. It combines computer-vision pothole detection with live or CSV-sourced accelerometer telemetry to produce certified infrastructure audit reports, tiered PWD repair-cost estimates, and a predictive Tire Wear Particle (TWP) environmental model — all in a single zero-dependency web application.

The project was built between 22 May and 31 May 2026 (nine days) by our team. Given that timeline, we used Roboflow's hosted CV inference API rather than training a model from scratch. The hackathon guidelines explicitly permit third-party tools; this is documented in the README and throughout this manual. Training a custom model tuned to Indian road conditions is a planned next step.

Why Tire Wear Particles?

The TWP environmental modeling component was directly motivated by a review paper written by one of our team members before this hackathon, which surveyed the literature on non-exhaust vehicle emissions and flagged tire-tread microplastic pollution as a seriously under-regulated hazard in Indian urban contexts. (OECD, TNO, Emissions Analytics — see Section 5).

Review paper link: [IJIRT191521_PAPER.pdf](#)

1. System Architecture

TreadGuard CV is a single-page web app in vanilla JavaScript (ES2022, strict mode) — no framework, no build system, no transpilation. The client is three files: index.html, styles.css, and app.js. A lightweight Node.js proxy (server.js) handles the Roboflow API call and keeps the API key server-side.

1.1 Data Flow

Stage	Detail
Input	Road video (MP4/WebM), still image, or live camera; optional G-force CSV
Frame capture	HTML5 Canvas drawImage → JPEG base64 at native resolution
AI inference	POST to /api/analyze (server.js proxy) → Roboflow Workflow API → bounding boxes + confidence
Human validation	Operator approves or rejects individual bounding boxes
Cost engine	Approved boxes → pixel-to-metre → tiered PWD rates → certified cost
TWP model	Vehicle mass + passengers + G-force → emission formula → mg/km
Output	Three downloadable HTML-to-PDF reports: AI Audit, Certified Manual Audit, TWP Environmental

1.2 Technology Stack

Component	Detail
Language	Vanilla JavaScript (ES2022) — no frameworks, no build step
AI inference	Roboflow Hosted Workflow API (REST/JSON)
Sensor input	Web DeviceMotion API (live) or CSV upload (recorded)
GPS	Web Geolocation API; Pune coordinates as offline fallback
Canvas	HTML5 Canvas 2D — bounding boxes, waveform, skid gauge, quality ring, mini-map
Reports	Blob-based HTML + window.print() → PDF via browser print dialog
Server proxy	Node.js (server.js) — holds API key, not in client bundle

2. Getting Started

2.1 Prerequisites

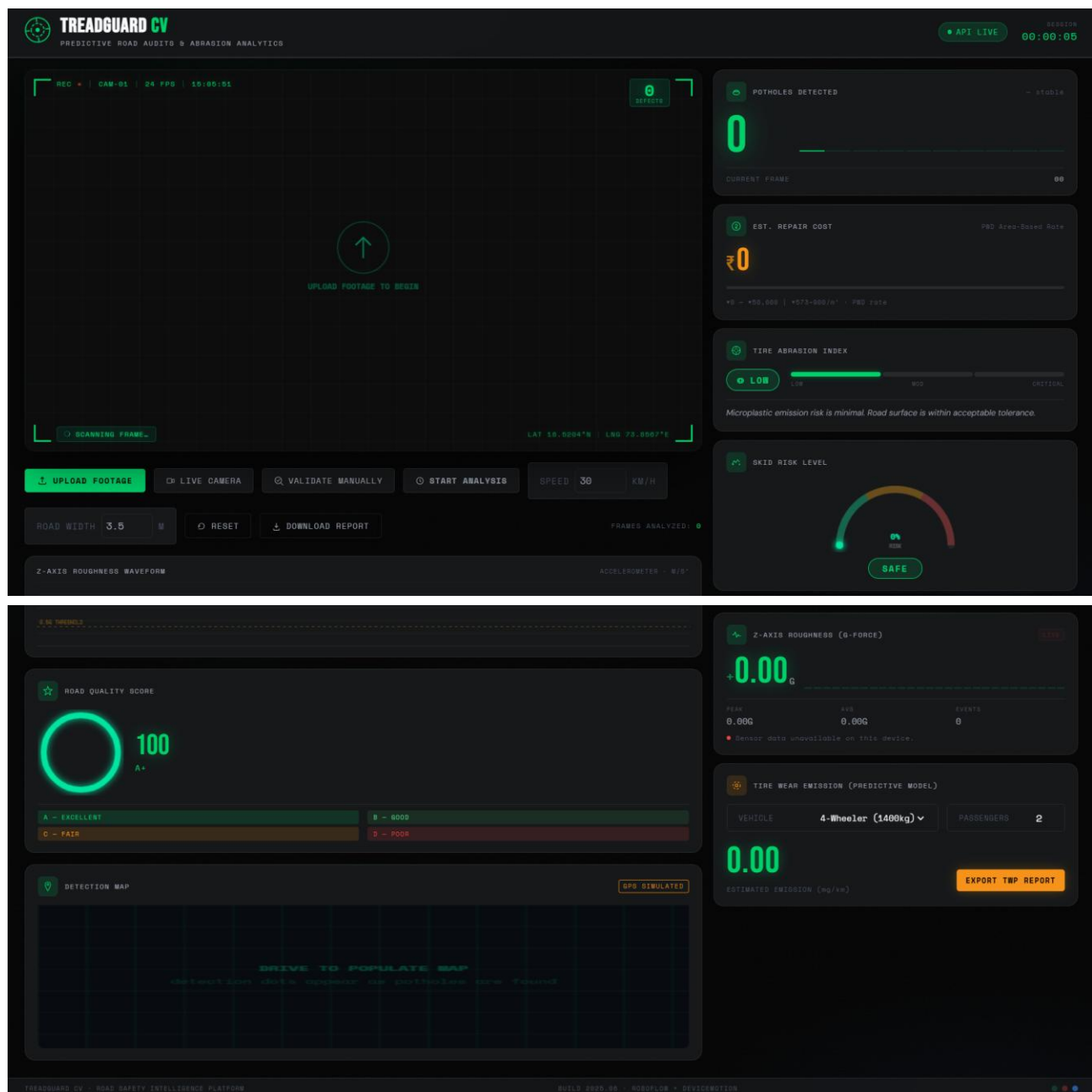
- Node.js 18+ (for the server.js proxy)
- A Roboflow account with a deployed Workflow and API key
- Chrome 112+, Firefox 120+, or Safari 16.4+
- HTTPS or localhost (required for DeviceMotion API on mobile)

2.2 File Structure & Setup

Project layout: treadguard-cv/index.html · styles.css · app.js · server.js · README.md · /trial-material

To run: (1) Open server.js and add your Roboflow API key and Workflow endpoint. (2) Run node server.js (defaults to port 3000). (3) Navigate to <http://localhost:3000>. The dashboard loads immediately — upload footage to begin.

server.js is only needed to proxy the Roboflow call and keep the key out of the client bundle. The rest runs entirely in the browser.



3. User Workflow

The operator follows five stages in sequence.

Stage 1 — Upload Footage & Telemetry

Click 'Upload Footage' and select a road video (MP4, WebM, MOV) and/or a G-force CSV recorded on the same drive. A confirmation modal shows the loaded filenames before analysis begins.

If no CSV is uploaded, the Z-axis panel shows 'No Telemetry Data'. On HTTPS mobile, live DeviceMotion populates the sensor panel automatically.

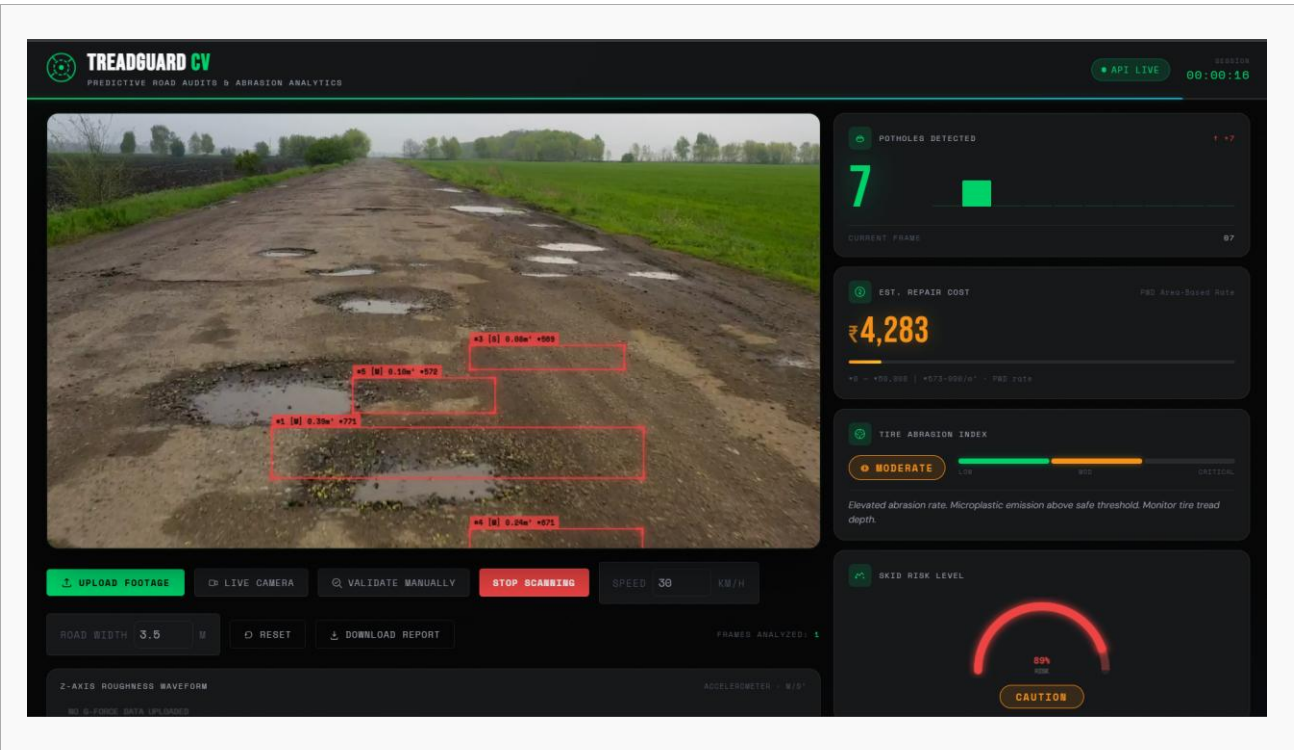
Stage 2 — Configure Calibration

- **Road Width (m):** real-world lane width used for pixel-to-metre conversion. Default 3.5 m (IRC:86 single lane).
- **Vehicle Speed (km/h):** governs the adaptive frame-sampling interval — faster speeds trigger more frequent captures.

Stage 3 — Run AI Scan

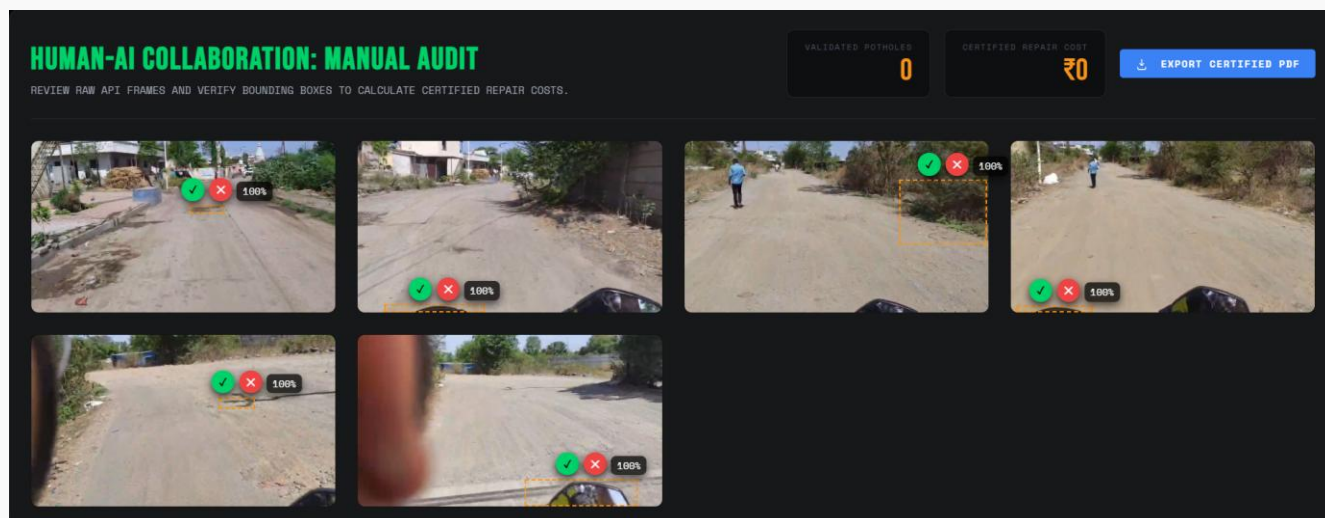
Click 'Start Analysis'. The system plays the video, captures frames at adaptive intervals (min 300 ms / max 2,500 ms), skips duplicates within 0.8 s, and sends each frame to Roboflow. Detections below 65% confidence or covering >60% of the frame are filtered out. All dashboard metrics update live — pothole count, repair cost, abrasion index, skid risk, quality score, TWP emission, and mini-map.

When the video ends, the system displays 'AWAITING HUMAN AUDIT' and scrolls to the validation dashboard.



Stage 4 — Human Validation

The validation dashboard shows every detected frame in reverse-chronological order, with annotated bounding boxes and a confidence badge. The operator ticks (✓) genuine potholes and crosses (X) false positives. Approved detections are added to the certified repair cost tally in real time. This is the core of the Human-AI Collaboration theme — the AI casts a wide net; the human applies judgment.



Stage 5 — Export Reports

Report	Button	Contents
AI Audit Report	Download Report	All AI-flagged potholes, repair costs, severity timeline, top-5 events, Z-axis stats
Certified Manual Report	Export Certified PDF	Human-validated potholes only. Marked 'Human Audited & Certified'. Suitable for PWD submission.
TWP Environmental Report	Export TWP Report	Tire wear emission estimate (mg/km), vehicle parameters, bump events, full formula with live values.

4. Spatial Calibration: Pixel-to-Metre Conversion

Roboflow returns bounding boxes in a fixed 640 × 360 pixel virtual frame regardless of video resolution. TreadGuard CV converts these to real-world metres using the known road lane width as a calibration anchor, assuming the lane fills the horizontal field of view:

metres_per_pixel = road_width_metres ÷ 640

Pothole area is then: width_m × height_m = (det.width × mpp) × (det.height × mpp). The default road width is 3.5 m (IRC:86 single lane). The Road Width field accepts 1.0–12.0 m to cover any road category.

4.1 IRC:86 Reference Widths

Road Type	Width
Single lane (default)	3.5 m
Two-lane road	7.0 m
Village road	3.0–3.75 m
Expressway lane	3.65–3.75 m

4.2 Known Limitations

- Zero camera pitch assumed. An angled mount will underestimate far-field pothole area.
- Full-width frame assumed. If the camera captures verge or sky, adjust the road width input accordingly.

- No perspective correction. Potholes near the top of the frame are slightly underestimated in area. Homography-based calibration is planned.

In practice, the tripwire accumulation mechanism, which only counts and costs a detection when its bounding box crosses the bottom 50% of the frame, acts as a passive perspective filter. Potholes are tallied at the point in the frame closest to the vehicle, where the camera angle introduces the least distortion. Far-field detections in the upper half, which carry the highest area error, are displayed on the HUD but excluded from the cost tally until they migrate into the near-field zone. This does not eliminate perspective error but substantially reduces worst-case area overestimation from far-field detections.

Trial videos were recorded at handlebar height on a two-wheeler, with the phone mounted approximately perpendicular to the road surface (dashcam-style). Road width was set to 3.5 m throughout.

5. PWD Repair Cost Model

Rather than a flat per-pothole rate, TreadGuard CV uses a tiered, area-based model grounded in PWD Schedule of Rates. Larger potholes cost proportionally more; the estimate is derived from the actual pixel area of each bounding box converted to m².

Cost per pothole = (area_m² × rate/m²) + ₹500 mobilisation

The ₹500 flat mobilisation charge reflects labour call-out and equipment setup — consistent with PWD SOR line items for site mobilisation.

Category	Area Range	Rate (₹/m ²)	Repair Type
Small (S)	< 0.10 m ²	₹900	Micro-patch / crack seal
Medium (M)	0.10–0.50 m ²	₹700	Cut and fill, bituminous macadam
Large (L)	> 0.50 m ²	₹573	Full-depth reclamation or dense overlay (PWD SOR base rate)

Bounding box labels show the size category, area, and cost. Example: #1 [M] 0.14m² ₹1,098 means detection 1, medium pothole, area 0.14 m², cost = (0.14 × 700) + 500.

PWD rates vary by state and revision cycle. These figures are representative estimates — verify against the current local SOR for official procurement.

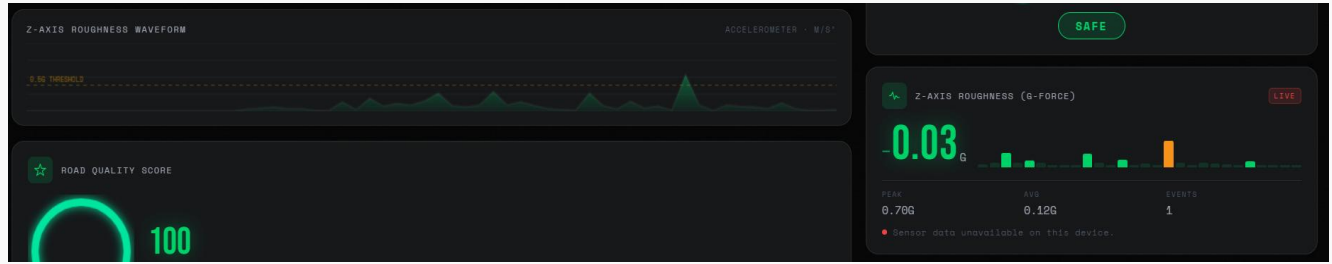
6. Z-Axis Accelerometer & Road Roughness

TreadGuard CV accepts Z-axis (vertical) accelerometer data from two sources: live DeviceMotion API (when opened on mobile over HTTPS), or a CSV from any smartphone accelerometer app (Physics Toolbox, PhyPhox, etc.). The CSV parser identifies the Z-axis column by header name matching.

When a CSV is paired with a video, sensor readings are synchronised to playback position using a progress-ratio mapping: `csv_index = floor( (currentTime / duration) × row_count )`.

6.1 Derived Metrics & Classification

Metric	Definition / Classification
Current G	Most recent Z-axis reading (absolute value, in G)
Peak G	Maximum absolute Z reading across the session
Avg G	Running mean of last 60 samples (~30 s at 2 Hz)
Bump Events	Count of readings where G > 0.5G — each treated as a significant road impact
G < 0.4 — Normal	Green readout; low abrasion; safe skid risk
0.4–0.8 G — Elevated	Amber readout; moderate abrasion; caution
G > 0.8 — Critical	Red readout; critical abrasion; hazardous skid



7. Tire Wear Particle (TWP) Predictive Model

Tire and road wear particles (TRWP) are a leading source of microplastic pollution in urban runoff. Unlike exhaust emissions, TRWP are generated continuously and scale with surface roughness, vehicle mass, and road impacts. TreadGuard CV's TWP module quantifies this in real time, extending the platform from a simple pothole mapper to an environmental telemetry tool.

7.1 Emission Formula

$$E_{twp} = (W_{veh} + N_{pax} \times 68) \times \beta \times (1 + G_{avg} \times 1.5) \times (1 + E_{bump} \times 0.02)$$

Symbol	Meaning
E_{twp}	Predicted tire wear emission (mg/km)
W_{veh}	Base vehicle kerb weight (kg)
N_{pax}	Passenger count
68	Average adult mass (kg) — WHO/ICMR South Asian reference
$\beta = 0.067$	Base friction coefficient (mg/km per kg) — midpoint of Kole et al. & Panko et al.
G_{avg}	Session average Z-axis G-force
1.5	Roughness sensitivity coefficient (empirical)
E_{bump}	Count of bump events ($ G > 0.5G$)
0.02	Impact scaling factor — 2% additional emission per event

7.2 Coefficient Derivation

$\beta = 0.067$ mg/km/kg is the midpoint of Kole et al. (2017) and Panko et al. (2019) central estimates for mass-normalised TRWP emission on smooth road surfaces. The roughness multiplier $(1 + G_{avg} \times 1.5)$ yields 1.6× at $G_{avg} = 0.4G$ and 2.2× at $0.8G$ — both within the Emissions Analytics (2023) observed range of 1.4–2.0× on rough versus smooth roads. The impact multiplier adds 2% per bump event; at 30 events it contributes an additional 1.60× factor, consistent with literature on pothole-driven abrasion spikes.

7.3 Vehicle Configurations

Category	Kerb Weight	Warning Threshold	Critical Threshold
2-Wheeler	150 kg	> 150 mg/km	> 250 mg/km
4-Wheeler (default)	1,400 kg	> 150 mg/km	> 250 mg/km
Heavy Duty Transit	4,500 kg	> 450 mg/km	> 600 mg/km

8. Roboflow AI Integration

Roboflow's hosted Workflow API is the computer-vision inference engine. It was chosen because: (1) the nine-day build window ruled out training a custom model; (2) Roboflow's pre-trained models cover diverse road types; (3) its REST API returns clean structured JSON with no ML infrastructure overhead; and (4) its use is explicitly permitted under hackathon rules and documented here.

8.1 API Architecture

- Browser captures a video frame as JPEG base64.
- Browser POSTs to /api/analyze on server.js (server-side proxy).
- server.js attaches the API key and forwards to the Roboflow Workflow endpoint.
- Roboflow returns JSON with bounding box coordinates and confidence scores.
- app.js parses, filters ($\geq 65\%$ confidence, $< 60\%$ frame-fill), and renders.

The multi-format parser handles three Roboflow response structures depending on Workflow configuration. If a non-zero count is returned but bounding boxes are empty, placeholder boxes are synthesised and logged to the console.

8.2 Adaptive Frame Sampling

Instead of a fixed interval, the system estimates when a detected pothole will reach the bottom-third 'tripwire' zone and schedules the next capture at 60% of that approach time. The delay is clamped to 300–2,500 ms, preventing API flooding while ensuring fast-moving potholes are not missed.

Planned: fine-tune a YOLOv8 / RT-DETR model on curated Indian road imagery (Pune, Chennai, Mumbai conditions) to replace the hosted Roboflow API with an on-device pipeline.

9. Dashboard Components

Refer to the Dashboard screenshots pasted in section 2 – pg no. 3

Component	Description
Media Viewer & HUD	Main video/camera feed with transparent canvas overlay for bounding boxes, REC indicator, FPS counter, GPS coordinates, and detection badge.
Pothole Counter	Session total (tripwire accumulation, 1,200 ms cooldown to avoid double-counting) and current-frame count.
Repair Cost Card	Cumulative certified cost with a proportional bar (₹0–₹50,000).
Tire Abrasion Index	Three-level severity (Low/Moderate/Critical) driven by rolling average of detected area per frame $\times 20$.
Skid Risk Gauge	Semi-circular canvas gauge (0–100%); green 0–33%, amber 33–66%, red 66–100%.
Road Quality Score	0–100 score: $\max(0, 100 - \text{severity} \times 7)$. Graded A+ (≥ 90) through F (< 20). Shown in a circular ring chart.
Z-Axis Panel	Live G-value, 28-bar history chart, peak/avg/bump-event stats, roughness waveform.
TWP Panel	Vehicle type + passenger count inputs; predicted emission output updated on every Z reading.
Detection Mini-Map	400 $\times$ 200 canvas; geo-coded detection dots (green < 3 , amber 3–7, red > 7 per event). Falls back to Pune centroid when GPS is unavailable.

10. GPS & Live Camera

TreadGuard CV calls `navigator.geolocation.watchPosition()` on startup. If permission is granted, real coordinates feed the HUD and detection log. If denied, the system falls back to Pune centroid (18.5204°N, 73.8567°E) and marks the GPS badge amber as 'GPS SIMULATED' — the map and reports are always populated with a geographically plausible value.

Live camera mode requests `getUserMedia()` for the rear camera and runs the same adaptive scan pipeline as a video file. On a phone over HTTPS, both GPS and DeviceMotion are live simultaneously, so all three reports reflect real-time data with no CSV needed.

Live camera and DeviceMotion require HTTPS. `getUserMedia()` will be blocked on plain `http://localhost`.

11. Trial Material & Test Results

Trial material was recorded during test drives in Pune, Maharashtra on 29 May 2026 and is available in the Google Drive folder on the cover page.

Video	CSV Pair	Conditions
trial video 1.mp4	trial video 1 accelerometer.csv	Hilly Area Road - Slope
trial video 2.mp4	trial video 2 accelerometer.csv	Mixed condition, includes unpaved section
trial video 3.mp4	trial video 3 accelerometer.csv	Dusty rural road, multiple detections, used for the included reports

Three sample reports from trial video 3 are included: AI audit, Manual Audit & TWP Report

The Roboflow model was trained mostly on Western road surfaces. On Indian dusty, unpaved roads it occasionally flags shadows and dark soil patches at high confidence. This is expected and is exactly why the human validation step exists. An operator rejects these in one click, and the certified report reflects only the approved detections.

Improvement priority: fine-tune on Indian road imagery to reduce false positives on dusty bare-earth surfaces.

12. Known Limitations & Planned Work

Current Limitations

- Custom AI model: the Roboflow-hosted detector would benefit from fine-tuning on Indian road imagery.
- Perspective distortion: no correction for camera angle or far-field foreshortening.
- GPS simulation: desktop browsers fall back to Pune centroid; real GPS needs a mobile device on HTTPS.
- Single-lane calibration: one fixed road width per video. Variable-width roads need per-frame calibration.
- TWP model: linear approximation; a full physics model would add tyre compound, temperature, and lateral forces.

Planned Improvements

- Train and deploy a custom YOLOv8 model on curated Indian road defect datasets.
- Homography-based perspective calibration using vanishing-point estimation.
- OpenStreetMap tile rendering in the mini-map for real geographic context.
- Route export (GeoJSON) for GIS integration with PWD mapping systems.
- Expanded TWP model with compound-specific emission factors and temperature correction.
- Multi-language UI support (Hindi, Tamil, Marathi) for field deployment.

13. References & Standards

Road Engineering

- IRC:86 — Geometric Design Standards for Urban Roads in Plains. Indian Roads Congress.
- IRC:SP:16 — Guidelines for Surface Evenness of Highway Pavements. Indian Roads Congress.
- MoRTH Specifications for Road and Bridge Works, 5th Revision. Ministry of Road Transport and Highways.

PWD Schedule of Rates

- Public Works Department Schedule of Rates — Bituminous Works. (State-specific; verify against current local SOR for procurement.)

Tire Wear Particle / TRWP

- Continental AG (2022). Tire & Road Wear Particles — Causes, Consequences, Solutions.
- Kole, P.J. et al. (2017). Wear and Tear of Tyres. Int. J. Environmental Research and Public Health, 14(10), 1265. <https://doi.org/10.3390/ijerph14101265>
- Panko, J.M. et al. (2019). Occurrence of tire rubber-derived contaminants in watershed systems. Chemosphere, 230, 474–482.
- Emissions Analytics (2023). TRWP Real-World Emission Measurements — Field Test Series. Technical Report.
- OECD (2021). Towards Eliminating Plastic Pollution by 2040. OECD Publishing, Paris.
- IJIRT (2026). Framework for Environmental Impact Assessment of Road Wear Microplastics in Urban Indian Contexts. Vol. 13.

Computer Vision & AI

- Roboflow Workflows Documentation. <https://docs.roboflow.com/workflows>
- Jocher, G. et al. (2023). YOLOv8 by Ultralytics. <https://github.com/ultralytics/ultralytics>

Review Paper (Team member-authored)

TWP motivation review (pre-hackathon): [IJIRT191521_PAPER.pdf](#)
